From Optimal Control to Diffusion Policy

Behavioural Cloning vs. Diffusion Policy on Double-Pendulum Swing-Up & Cube Stacking

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Group Project TN2

June 29, 2026

Outline

Introduction

Goal of the project

Question

Compare two imitation-learning control methods on the same data.

BC (baseline)

Regress the expert action directly from the state — a single deterministic map.

Diffusion Policy

Generate an action *chunk* from noise by iterative denoising, conditioned on the state.

Two tasks

- Double-pendulum swing-up (small, fast, unstable balance task)
- Robotic cube stacking — Stack (2 cubes) and StackThree (3 cubes)

Same demonstrations per task $\Rightarrow$ the comparison reflects the *method*, not the data.

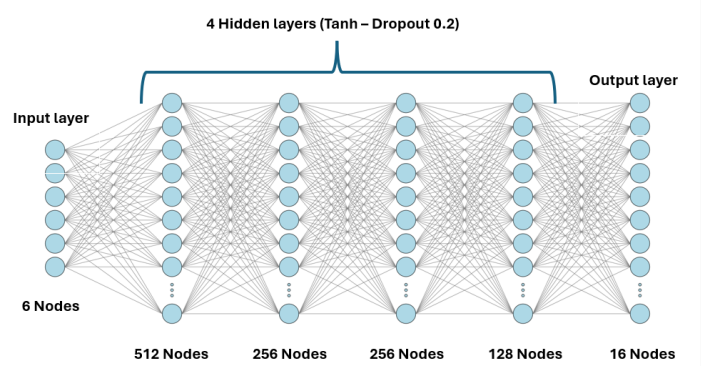
Part I — Double-Pendulum Swing-Up

Part I — roadmap

1. **BC model** — network & training
2. **Diffusion background** — forward / reverse process
3. **Generating the expert dataset** — TrajOpt $\rightarrow$ TVLQR $\rightarrow$ rollout
4. **Diffusion Policy model** — conditioning, network, control tricks
5. **Comparison** — metrics, robustness, ablations

1. Behavioural Cloning - Network Architecture

- Input: feature transform $\phi(s_t) \in \mathbb{R}^6$
- 4 hidden fully-connected layers, tanh, dropout with probability 0.2
- Output 16 $\rightarrow$ reshaped to 8×2 torques

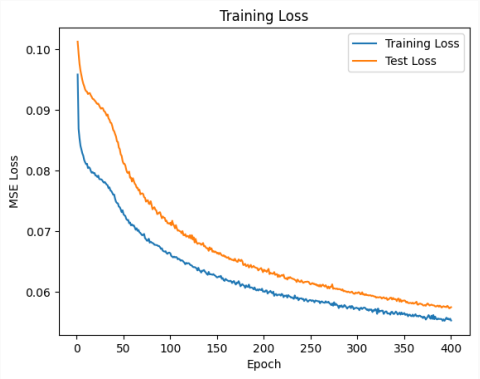


BC Model architecture.

2. BC - Training

Learn $\pi = (a_{t:t+7} \mid s_t)$ — an 8-step action chunk.

- Wrap-safe angular features $\phi = [\sin q_1, \cos q_1, \ldots, \tilde{q}_2] \in \mathbb{R}^6$
- Min–max normalise to $[-1, 1]$ actions and velocities (stats saved)
- 90/10 train/val
- MSE loss, Adam 10^{-3}, early stop (no overfit)
- 512 batch size
- 400 epochs



Train/test loss — stopped early

2. Diffusion models — background

Forward (noising):

$$x_t = \sqrt{\bar{\alpha}_t}\, x_0 + \sqrt{1-\bar{\alpha}_t}\, \epsilon, \qquad \epsilon \sim \mathcal{N}(0, I)$$

Reverse (denoising): a network $\epsilon_\theta(x_t, t, c)$ predicts the added noise.

Training (ϵ-MSE)

- Sample t, add noise to get x_t
- Minimise $\|\epsilon_\theta(x_t, t, c) - \epsilon\|_2^2$

Sampling (control time)

- Start from pure noise, iterate $T \to 1$
- Deterministic: drop ancestral noise $\Rightarrow$ posterior mean, low variance

Formulation follows *Understanding Deep Learning* (Prince, 2023).

2. DDPM sampler — double pendulum

Sample an action chunk by ancestral denoising (DDPM): start from pure noise $a_T \sim \mathcal{N}(0, I)$ and iterate $t : T \to 1$ with $T = 20$.

One reverse step

$$a_{t-1} = \frac{1}{\sqrt{1-\beta_t}}\Big(a_t - \frac{\beta_t}{\sqrt{1-\bar{\alpha}_t}}\,\epsilon_\theta(a_t, t, c)\Big) + \sqrt{\beta_t}\,z$$

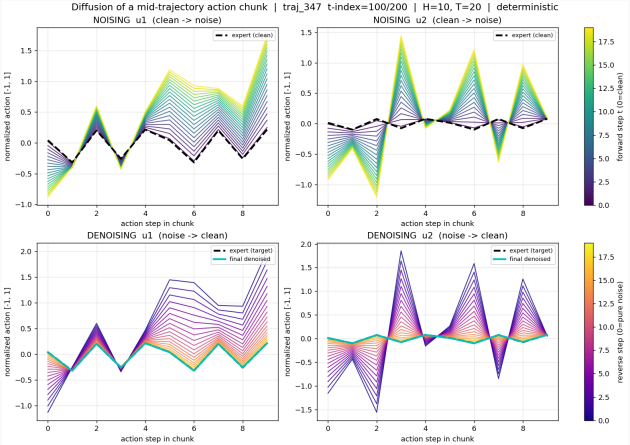
- Linear β-schedule ($10^{-4} \to 0.02$), EMA 0.999
- *Markovian*: every step removes a little noise, re-adds a little ($z \sim \mathcal{N}(0, I)$)

Control tweak: drop z

Set ancestral noise $z = 0 \Rightarrow$ posterior mean each step. Low action variance is what lets it *hold* the unstable upright — stray torque noise topples the pendulum.

All $T = 20$ steps run every re-plan (n_{exec}=1).

2. DDPM in action — noising & denoising a chunk



One mid-rollout expert chunk (H=10, two torques u_1, u_2). **Top:** forward process buries the expert (dashed) in noise; **bottom:** the deterministic reverse process recovers it almost exactly (cyan $\approx$ dashed, reconstruction MSE $\sim 10^{-6}$).

3. Generating the expert dataset

Classical optimal-control pipeline — every sample dynamically valid by construction.

1. **Dynamics model** — planar 2-link EoM, all parameters read from `dp.xml`; smooth tanh friction (differentiable).
2. **Trajectory optimisation** — direct collocation NLP solved with IPOPT (JAX autodiff); swing $[0, 0, 0, 0] \to [\pi, 0, 0, 0]$.
3. **TVLQR tracking** — $u_t = u_t^* - K_t(x_t - x_t^*)$; nearest-neighbour re-lock for recovery.
4. **Closed-loop rollout** in MuJoCo @ 20 Hz, logging (state, torque).

DAgger-style coverage

Perturb *applied* torque ($\sigma = 0.15\tau_{\max}$), log the *clean* command $\Rightarrow$ teaches recovery from off-nominal states. Keep only successful rollouts ($\sim$600 trajectories).

4. Diffusion Policy model (Chi et al. 2023)

Denoise an action chunk $a^{1:H}$ ($H = 10$), conditioned on a clean context c.

Representation

- Wrap safe angular features ϕ
- Min–max normalise to $[-1, 1]$ actions and velocities (stats saved)
- c = last K=4 states + goal $\in \mathbb{R}^{30}$

Network ϵ_θ

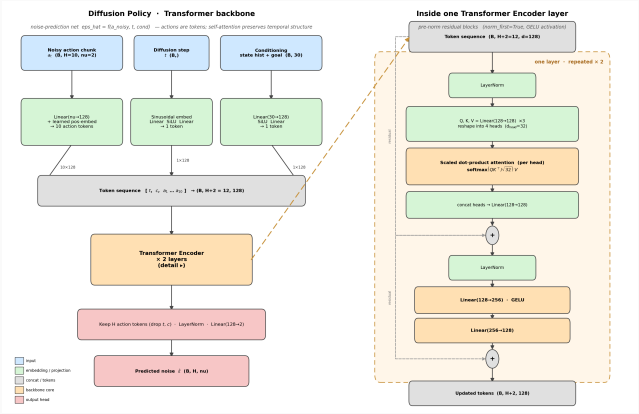
- Trajectory Transformer (default) vs. MLP
- Linear β-schedule, $T = 20$ steps, EMA 0.999

What makes the loop stable

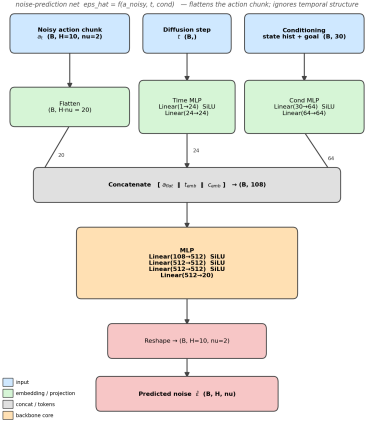
- Deterministic (noise-free) sampling
- **Single-step execution** (n_{exec}=1): re-plan every step

Goal token is effectively constant (one upright goal) ⇒ not truly conditionable.

4. Denoiser networks — Transformer vs. MLP

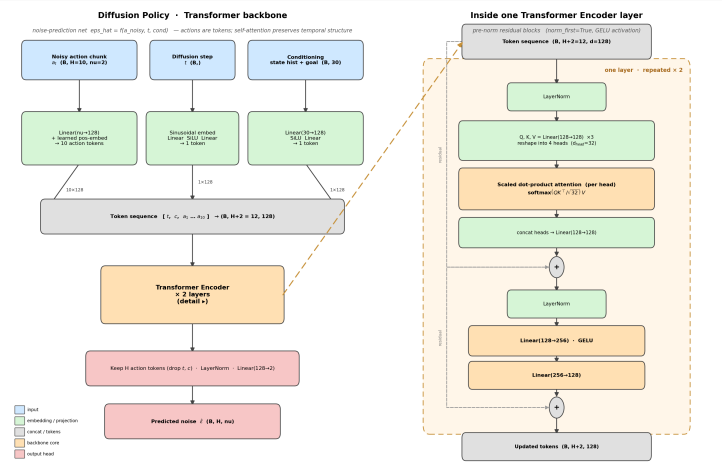


Trajectory Transformer denoiser ϵ_θ (default).



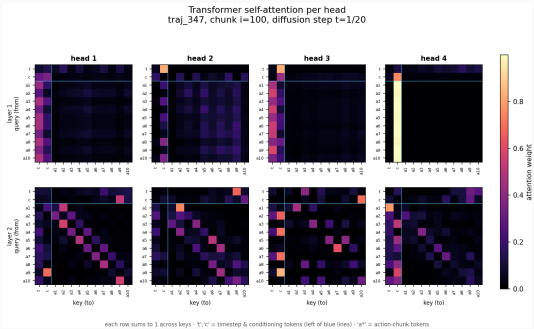
MLP denoiser (ablation baseline).

4. Trajectory Transformer denoiser

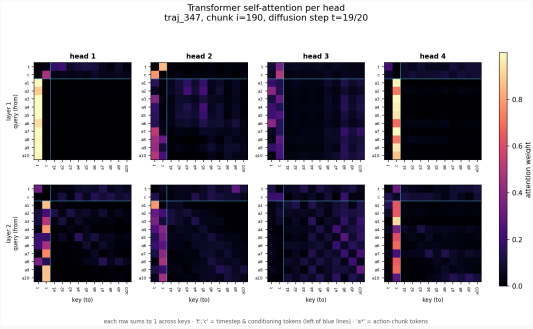


Tokens: diffusion timestep `t` + conditioning `c` + action-chunk `a1...a10`, mixed by self-attention.

Attention heads — early vs. late denoising



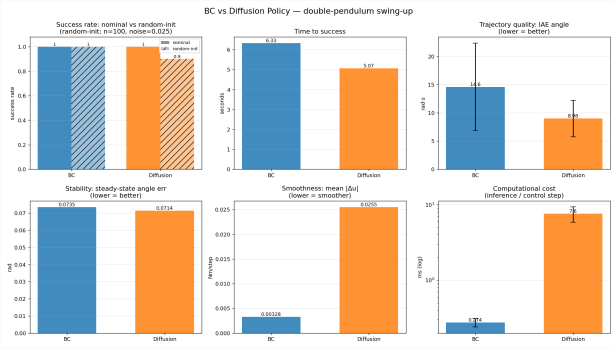
Early denoising (t=1/20): action tokens still attend to each other.



Late denoising (t=19/20): mass collapses onto the c (conditioning) column.

Rows sum to 1; `t`,`c` = timestep & conditioning tokens (left of blue lines), `a*` = action-chunk tokens.

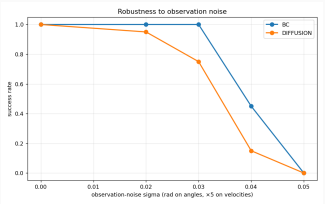
5. Comparison — aggregate metrics



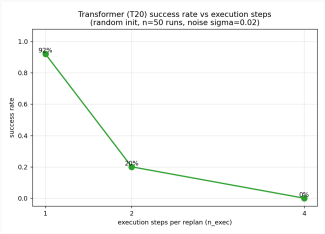
Both swing up: **BC** 100%, **Diffusion** 90% (random init).

- **Diffusion** swings up *faster* (5.07 vs 6.33 s) and lower IAE ($\sim$40%)
- **BC** holds upright far smoother ($8\times$ less $|\Delta u|$)
- **BC** $\sim 28\times$ cheaper per step (0.27 vs 7.6 ms)

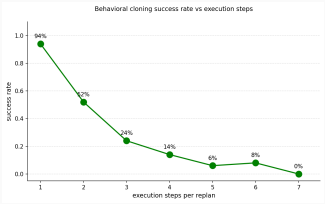
5. Comparison — robustness & ablations



BC degrades more gracefully under observation noise.



Committing to > 1 open-loop action destroys performance.



BC performs also performs best when open-loop actions are less than 1.

Design choices are not free parameters: transformer backbone (MLP collapses to 2%) and $n_{\text{exec}} = 1$ are the difference between a working and a broken controller.

Part I — takeaway

On the double pendulum, BC is the better controller

Both succeed, but BC holds the upright smoother, is more noise-robust, and $\sim 28\times$ cheaper. The diffusion policy is fragile (needs per-step replanning, chatters near the unstable equilibrium).

Diffusion's edge — multimodal, temporally-correlated actions — does not pay off on a single, unimodal balance task.

Part II — Robotic Cube Stacking

Task & control setup

Franka Emika Panda (7 DoF) in MuJoCo / robosuite, task-space OSC_POSE control.

Action space (7-D)

- Translation $\Delta x, \Delta y, \Delta z$
- Rotation delta (axis-angle)
- Gripper state $\in [-1, 1]$

State space

- Stack: **32-D** (2 cubes)
- StackThree: **48-D** (3 cubes)
- cube poses, gripper-cube vectors, eef pose

Goal: learn $\pi(a_{t:t+H} \mid s_t)$, receding-horizon control (H-step chunks).

Dataset — MimicGen

- Source: $\sim$10 human demos per task $\rightarrow$ MimicGen auto-generates large datasets ($D_0, D_1, \dots$).
- Four datasets used: `stack_d0/d1`, `stack_three_d0/d1` — 1000 trajectories each.
- HDF5 / robomimic format; only state-relevant keys used (no images).

Why it matters

Demonstrations contain *several valid ways* to finish a stack $\Rightarrow$ a multimodal action distribution — the regime where diffusion is expected to shine.

Models

BC baseline (MLP)

- 5 linear layers ($256 \to 512$), LayerNorm, ReLU, dropout 0.1
- State $\to$ 56 values $\to$ $(8, 7)$ horizon
- MSE loss, Adam 10^{-3}, 50 epochs

Diffusion Policy (Transformer)

- `TransformerDenoiser`: 4 layers, d=256, 4 heads
- State + timestep tokens prepended to H=8 action tokens
- ϵ-MSE, AdamW, K=100 steps, EMA 0.995

Testing: 50 episodes per model/task, control @ 20 Hz, horizon 500 steps; success = cubes stacked.

DDIM sampler — cube stacking

Same ϵ_θ trained over $K = 100$ noise levels, but sampled with the deterministic, non-Markovian DDIM update — letting us *skip* timesteps.

One reverse step — predict the clean chunk, then re-noise to level $k-1$:

$$\hat{a}_0 = \frac{a_k - \sqrt{1-\bar{\alpha}_k}\,\epsilon_\theta(a_k, k, s_t)}{\sqrt{\bar{\alpha}_k}}$$

$$a_{k-1} = \sqrt{\bar{\alpha}_{k-1}}\,\hat{a}_0 + \sqrt{1-\bar{\alpha}_{k-1}}\,\epsilon_\theta$$

No injected $z \Rightarrow$ the noise→action map is deterministic.

Why DDIM here

- Sub-sample 100→50 steps: 2× fewer network calls, ∼same quality
- Deterministic ⇒ reproducible, smoother chunks (one mode per noise seed)
- Still multimodal across seeds — what the long-horizon stack needs

H=8 receding horizon, EMA 0.995.

Evaluation — metrics & results

Metrics per planning step: trajectory **jitter** J, **path effort** E, **joint cost** C; plus **success rate**.

Task	Best success	BC	Diffusion
Stack	(H=8/4)	93%	**98–100%**
StackThree	(H=8)	56%	**70%**
Replan @ H=2 (StackThree)		1%	**36%**
Jitter (H=8, Stack)		**0.007**	0.022

- **Diffusion** stacks more reliably and tolerates more frequent replanning.
- **BC** is smoother (lower jitter) but completes fewer stacks — blurs the multiple valid motions.
- Both need the long H=8 horizon; H=1 breaks temporal coherence.

Conclusion

Conclusion

Neither method wins everywhere

The two tasks need *opposite* replanning rates, and each method suits one.

Double pendulum

(fast, unstable, replan every step)

- **BC** wins: smoother, robust, cheap
- Diffusion fragile, chatters

Cube stacking

(multimodal, long horizon)

- **Diffusion** wins: higher success
- BC averages valid motions $\rightarrow$ fails more

BC: simple, cheap control of small balance tasks. Diffusion: larger, multimodal manipulation.

Thank you

Questions?